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Unpacking the minimum viable product (MVP): a framework for use, goals and essential elements

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Abstract

Purpose – Entrepreneurial practice often relies on the minimum viable product (MVP) to test business model hypotheses, yet the conceptualization of its makeup remains inadequately defined, particularly in uncertain startup contexts. This paper aims to clarify how entrepreneurs should think about MVPs in terms of their uses, goals and essential components to enhance their effectiveness as a tool for navigating uncertainty.

Design/methodology/approach – Existing literature on MVPs is broad and often overlapping, leading to confusion regarding essential components and best practices for MVP development. Through a systematic analysis of existing published definitions, this paper offers clarity by proposing a framework that breaks the MVP concept up into process, goals and the fundamental elements necessary to launch an MVP. By doing so, it provides actionable guidance for entrepreneurs seeking to utilize MVPs in their business model testing endeavors.

Findings – This conceptual paper critically examines the lean startup approach, seeking to disentangle the complexities surrounding MVP development. Drawing on existing literature and practical insights, the study identifies and articulates a framework that clarifies the MVP concept along with the core elements required for creating a viable MVP, including (a) artistic elements, (b) a robust distribution channel and (c) an effective user feedback mechanism.

Originality/value – This paper contributes to theory development and pedagogical practices by providing a structured framework for understanding and implementing MVPs in entrepreneurial contexts. By identifying the minimal category elements of an MVP, it offers practical insights into entrepreneurs and educators alike, facilitating effective business model hypothesis testing in varied and uncertain environments.

Keywords Product development, Entrepreneurial education and training, Minimum viable product, Entrepreneur education

Paper type Conceptual paper

Introduction

The minimum viable product (MVP) is a foundational element of the lean startup methodology, which has gained widespread adoption among entrepreneurs, intrapreneurs, educators and students in the field of entrepreneurship. When startups need to test the assumptions embedded within their business models, they formulate hypotheses that require validation through real user interactions to reduce uncertainty (Blank, 2010). MVPs play a crucial role in this process by enabling startups to achieve this validation using the minimum resources necessary, thereby mitigating risk. Once these hypotheses are tested and assumptions validated or invalidated, startups can either pivot their business models or proceed toward building a more scalable version of the MVP to reach a broader customer base

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if the remaining risks and uncertainties are deemed manageable (Blank and Dorf, 2013; Ries, 2011).

Despite the popularity and frequent use of MVPs, a formalized understanding of what constitutes an MVP and how it can be systematically employed remains elusive. This is evident in the diverse definitions that have been proposed over time, each contributing to an evolving understanding of the concept but also adding layers of complexity that can confuse both practitioners and scholars (Shepherd and Patzelt, 2021). Given our review of the literature, we are left with practitioner and academic definitions along with practical examples. While there is certainly useful information across these understandings, we end up with considerable overlap, contrasting ideas and an overall lack of clarity that creates a less than useful situation for entrepreneurs, researchers and educators. Perhaps most importantly, the definitions offered are inconsistent in describing how to use an MVP, what an MVP should accomplish and what are the tangible elements of an MVP to accomplish the goal(s) that these definitions suggest. The most basic tenant of a useful definition should be clarity in understanding a concept, and something that many of these definitions struggle to provide is such clarity, especially when they are compared side-by-side. Further, as our review shows, existing definitions suppose that MVPs can be used in many ways, for many different purposes, without much clarity on what exactly goes into building an MVP.

Our criticism is not specifically a disagreement with what previous scholars and practitioners have proposed. While we do believe that some definitions we review are better than others, our contention is instead that they are largely incomplete when viewed as individual definitions and we would argue that *no* individual definition could fully encompass the full nature of what an MVP is, does and accomplishes. In our view, while MVPs themselves should be simple, the conundrum is that understanding what an MVP has become increasingly complex and hard to understand with standalone definitions.

Thus, our purpose is largely to answer the calls from Shepherd and Gruber (2021) for future research that looks to answer the many unanswered and unclear fundamental aspects that define an MVP. In doing so, we do not propose a new definition, but instead offer a framework to fully understand, envision and describe the composition, uses and accomplishments of MVPs. MVPs are an important part of new product/service development for entrepreneurs and intrapreneurs alike, but the vast array of definitions and implementations vary widely as discussed above. A clear framework for understanding MVPs can help entrepreneurs, developers, scholars and educators better conceptualize and apply the MVP concept.

The myriad of definitions and interpretations of MVPs has led to a fragmented understanding of their essential components and how they function as a replicable tool for entrepreneurs. This lack of clarity is problematic as it hampers both academic efforts to systematize knowledge and the practical ability of new entrepreneurs to design and deploy effective MVPs (Bruton *et al.*, 2023). Without a synthesized understanding of the core elements that constitute an MVP, there is a risk of misapplication or incomplete application of the concept in entrepreneurial practice.

The primary issue this paper addresses is the need for a clearer, more formalized understanding of MVPs that can guide scholars, practitioners and educators. We aim to provide a framework that goes beyond simple definitions, offering a deeper conceptualization of MVPs as "boundary objects" in the lean startup process (Shepherd and Patzelt, 2021). Through this lens, we argue that an MVP should function as an intermediary between the entrepreneur's vision and the feedback from a predefined group of stakeholders (e.g. customers, investors, employees) by following specific processes and utilizing specific elements toward the overarching goal of customer development. This framework not only clarifies the essential elements of an MVP but also enhances its utility as a tool for iterative learning and innovation.

In the following sections, we first review the literature related to MVPs, focusing on an extensive review of previously published definitions from academic journals and practitioner books. We then analyze these definitions to identify common themes that form the baseline for

our framework, specifically emphasizing that MVPs follow a structured process to achieve specific goals and are composed of necessary elements. We develop arguments for these essential elements, categorized into art, distribution channels and user feedback mechanisms and provide a comprehensive list of options within each category. Finally, we discuss the implications of our framework for future research, practical MVP design and pedagogy.

Literature review

The initial conceptions of MVPs are deeply rooted in both the lean launch and design thinking literature (e.g. Aulet and Ursache, 2013; Denning and Dunham, 2012; Kim and Mauborgne, 2015; Knapp *et al*., 2016; Maurya, 2012; Rose, 2016; Schlesinger and Kiefer, 2012; Taulbert and Schoeniger, 2010). Design thinking, which emerged in the mid-nineteenth century, initially focused on fields such as architecture and engineering by integrating John Dewey's conceptualization of experiential learning and William Morris' emphasis on the societal effects of design and the paramount importance of user needs. Over time, design thinking evolved into a broader field of technological research (e.g. Buchanan, 1992) and later made its way into the management domain (e.g. Martin, 2009). Many of the underlying principles of design thinking—such as experiential learning, customer focus and problem-solving—have since been integrated into entrepreneurship literature and practice, particularly within the lean launch framework (e.g. Osterwalder *et al*., 2014). This iterative process is now recognized as an instrumental and highly efficient method for startups to use MVPs as a guide for determining when to pivot and whether existing hypotheses about components of a given business model are validated or invalidated (Lortie *et al*., 2022).

As the concept of design thinking evolved, it began to intersect more directly with the field of entrepreneurship, particularly in the development of lean startup methodologies. The MVP, as a manifestation of these principles, was designed to embody the iterative and user-focused aspects of design thinking. However, while MVPs have been widely adopted, there remains a conceptual gap in fully understanding their limitations and the specific contexts in which they are most effective. For example, while MVPs are effective in reducing uncertainty and validating business model assumptions, there are scenarios where they might not fully address the complexities of certain entrepreneurial ventures, particularly those involving high levels of innovation or rapidly changing market conditions.

As research has evolved, more definitions have been published that have expanded the concept of MVPs, as presented in Table 1. To identify and compile definitions of MVPs from both academic journals and respected practitioner books, we conducted a comprehensive search using several academic databases, including Scopus, Web of Science, Google Scholar, IEEE Xplore and JSTOR. The search employed keywords such as "Minimum Viable Product," "MVP," "Lean Startup," "Design Thinking" and "Product Development" with Boolean operators and truncation to capture variations in terminology. Our search included all years of publication through 2022. We applied strict inclusion criteria, focusing on peer-reviewed articles and practitioner books that provided explicit definitions or discussions of MVP. Following the initial search, we utilized a snowball citation method to broaden our findings. This involved backward citation tracking to explore references within the core literature and forward citation tracking to identify subsequent works that cited the core articles. This iterative approach allowed us to uncover a wide range of MVP definitions, ensuring a comprehensive understanding of the term as it is conceptualized across different domains. The definitions were then systematically analyzed to identify common themes and variations, providing a nuanced view of the MVP concept in the literature.

For practitioners, Ries (2011) suggests that an MVP is a version of a new product, which allows a team to collect the maximum amount of validated learning about customers with the least effort. Therefore, it is an iterative process of idea generation, prototyping, presentation, data collection, analysis and learning. Whereas Blank (2005) states that an MVP has just those features (and no more) that allow a product to be deployed. Moogk (2013) offers that an MVP

Table 1. Existing definitions of MVP

Year	MVP definition	Author(s)	Process-based	Goal-based	Element-based	Elements named
2001	MVP is not a minimal product, it is a strategy and process directed toward making and selling a product to customers	Robinson	X			
2005	An MVP has just those features (and no more) that allow the product to be deployed	Blank			X	Minimum features
2011	MVP is a version of a new product, which allows a team to collect the maximum amount of validated learning about customers with the least effort. It is an iterative process of idea generation, prototyping, presentation, data collection, analysis and learning	Ries	X	X		
2012	An MVP is an offer that generates revenue for the company and that motivates customers to provide feedback and recommend it to other potential customers	Poole	X	X		
2013	An MVP is typically the first version of a product released to customers, and should contain only the absolute minimum in terms of features and design for it to become viable to the customer	Björk *et al.*			X	Absolute minimum
2013	An MVP will contain only the critical features of an envisioned product and is designed to test a specific hypothesis quickly	Blank and Dorf	X		X	Critical features
2013	MVP is a product with a minimum feature set targeting market opportunities and validates its value and growth hypotheses as soon as possible	Kline *et al.*	X		X	Minimum feature set
2013	An MVP is a product with a minimum feature set targeting market opportunities while profitably solving customer pain points	Moogk	X	X	X	Minimum feature set
2014	The intent is to move rapidly and frugally to a point where you can test your product and its fit against the latent market needs as well as understand whether your original business concept can make money at the scale envisioned or whether it requires changes in direction, strategy, or technology	Divan	X	X		
2014	MVP is a set of "minimal requirements," which meet the needs of the core group of early adopters or users	York and Danes			X	Minimal requirements
2014	MVP starts the learning and building process quickly. It allows the start-up team to collect as maximum validated learning about customers with least effort. The goal is to test the fundamental business hypothesis. It is not meant to be perfect	Miski	X	X		

(*continued*)

Table 1. Continued

Year	MVP definition	Author(s)	Process-based	Goal-based	Element-based	Elements named
2014b	MVP aims at defining the smallest possible implementation that brings added value to customers	Järvinen *et al.*		X	X	Smallest implementation
2014a, b	MVP is a product capable of being deployed to a subset of customers for evaluation	Read *et al.*	X			
2014	MVP is an experimental object that allows for empirical testing of value hypotheses	Münch *et al.*	X		X	Experimental object
2014	MVP is a key concept. The goal is to identify the most valuable features by iteratively experimenting the market	Mikkonen and Systä	X	X		
2014	The MVP model has data at its functional core: the develop-release-measure-iterate feedback loop drives the product, from its earliest state through to release, based on behavioral data harvested from users, not product team intuition	Seufert	X			
2014	The product is a set of "minimal requirements," which meet the needs of the core group of early adopters or users	York and Danes			X	Minimal requirements
2015a	MVP is used to test the business model by gathering and measuring customer feedback. Create a viable product with minimum effort. Requires the generation and testing of numerous ideas	Yli-Huumo *et al.*	X	X		
2015	MVP is a tool to collect customer feedback on the product in order to improve the product	Edison *et al.*	X			
2015	An MVP has just those features (and no more) that allow the product to be deployed and allow for reasoning with early adopters; some of whom will pay you money or give you feedback	Agostinho *et al.*		X	X	Minimum features
2016	In product development, the minimum viable product (MVP) is the product with the highest return on investment versus risk	Erder and Pureur		X		
2016	The least possible piece of functionality that makes sense . . .	Gousios			X	Least possible
2016a, b	MVP is a version of a new product with minimum features, which allows a team to collect the maximum amount of validated learning about customers with the least effort	Lenarduzzi and Taibi	X	X	X	Minimum features
2016	MVPs, defined as products with just enough features to gather validated learning about the products, is a major focus in early stages	Nguyen Duc and Abrahamsson		X	X	Just enough features

(*continued*)

Table 1. Continued

Year	MVP definition	Author(s)	Process-based	Goal-based	Element-based	Elements named
2016	MVP is used . . . to support validated learning, facilitate product design, bridge communication gaps and enable cost-effective development activities	Nguyen Duc and Abrahamsson	X	X		
2017	The MVP is a minimal and sufficient representation of new product idea or business hypothesis which allows starting the process of learning from customers through running experiments that can be validated scientifically	Saadatmand *et al.*	X	X		
2018	The concept of MVP . . . shows its adaptability in testing hypotheses about the target audience and guiding companies toward a stable business model	Hyrynsalmi *et al.*	X	X		
2019	The MVP concept from the lean startup methodology helps quickly start the learning process by integrating feedback from early adopters	Dennehy *et al.*	X			
2019	Artefacts used for the fast and quantitative market testing of a product or product feature	Ghezzi	X		X	Artefacts
2019	A representative proxy of the final product, to validate either new technology or to elicit customer requirements	Nguyen Duc *et al.*		X	X	Representative proxy
2021	An MVP . . . is the product whose goal is to test the fundamental business hypothesis. It is a version of a new product that allows a team to collect the maximum amount of validated learning about customers with the least effort	Lee and Geum	X	X		
2020	The MVP is a version of the product or service that is shared with customers, enabling the tracking of information that can be used to validate or invalidate assumptions and determining the new venture's next steps	Winkel *et al.*	X	X		
2021	MVPs are boundary objects (or a business model) that facilitate communication	Shepherd and Gruber	X		X	Boundary object
2022	MVPs in digital startups serve purposes like feature validation and idea extraction at different phases, helping to design better products with desired features and innovation	Umbreen *et al*	X	X		

Source(s): Authors' own work

is a product with a minimum feature set targeting market opportunities while profitably solving customer pain points. Whereas Poole (2012) states an MVP is an offer that generates revenue for the company and that motivates customers to provide feedback and recommend it to other potential customers.

Practitioners have also provided a roundabout way of communicating MVPs by using hypothetical and real-world examples of successful use cases. Table 2 illustrates different types of MVPs. Examples serve as a creative suggestion for MVP types that a startup would consider deploying but does not necessarily provide concrete guidelines or specific ideas about the individual elements that an entrepreneur may choose. Further, these examples are void of much direction in terms of the exact ways an entrepreneur might use them and for what goals they could accomplish. For instance, an explainer video and video MVP are ways a startup can build their MVP, but the definition for each of these MVP types lacks what content the entrepreneur requires for deployment. It does not clarify how an entrepreneur might get these MVPs in front of potential customers, the ways in which features can be highlighted, the style of the video or the steps in which to create one. Examples also commonly fail to provide any explanation about the context in which a specific type of MVP might be more or less appropriate or efficient. Furthermore, there are many overlapping examples that can make it difficult to determine which one is best for a specific business model an entrepreneur might be evaluating, potential iterations and/or testing the assumptions with the highest risk. Finally, as seen in practice, entrepreneurs might use combinations of these different examples of MVPs, and likely should, for better, quicker and more efficient testing.

In earlier academic literature, there is also considerable variation in terms of definitional conceptualizations. For example, Järvinen *et al.* (2014a, b) suggest an MVP aims at defining the smallest possible implementation that brings added value to customers. Similarly, Read *et al*. (2014a) conclude that an MVP is a product capable of being deployed to a subset of

Table 2. Practitioner MVP examples

MVP title	Description
Explainer video	A short animation that explains what your product does and why users should buy it. The video is typically simple and typically thirty seconds to a few minutes in length
Landing page	A landing page is used to quickly communicate the startup's proposals, to diffuse objections, and to call the visitors to action
Wizard of Oz	A user interface that looks like a real working product, but the actual business process is manually carried out. The purpose of this MVP is to demonstrate the complete job done by the product
Concierge MVP	A manual service that consists of exactly the same steps users would go through with the product
Piecemeal MVP	Similar to wizard of Oz MVP, however, execution of the tasks is completed using a variety of existing tools
Mockup MVP	Typically, a blueprint in the form of a paper or digital prototype or wireframe that is then used to represent the product user interface without any functionality
Public project proposal	Kickstarter and other rewards-based crowdfunding sites that allow for users to pre-purchase the product and provide a great way to raise funding for initial orders
Single feature MVP	A prototype that implements only the most important function of the product
Rip off MVP	A successful product is used to get feedback from users/customers followed by a pivot or improvement in a different direction
Video MVP	A video of a product, prototype or visions that demonstrates the proposed hypothesis being tested to the customer
Smoke tests	An illusion that the product is at a later stage than it really is, can be conceptualized in most of these other versions of MVPs, commonly used along with video MVP
Email MVP	Testing the value proposition via email, often optimal and most effective for a startup with a large existing customer base

Source(s): Authors' own work

customers for evaluation. More comprehensively, Miski (2014) stipulates that an MVP starts the learning and building process quickly and allows the start-up team to collect maximum validated learning about customers with the least effort. The goal is to test the fundamental business hypothesis rather than developing a perfected product. Similarly, Yli-Huumo *et al.* (2015a) suggest that an MVP is used to test the business model by gathering and measuring customer feedback, by creating a viable product with minimum effort that requires the generation and testing of numerous ideas.

More recent academic publications such as Shepherd and Gruber's (2021) explanation of the MVP as a version of a product that enables the build-measure-learn feedback loop using the minimal amount of resources (Ries, 2011), an MVP will contain only the critical features needed to test hypotheses (Blank, 2013a), and the purpose is to learn and test assumptions (Blank and Dorf, 2013). In terms of the relevant academic literature, Shepherd and Gruber (2021) rightfully look toward the design, design thinking and prototyping literature (e.g. Brown *et al.*, 2008; Luchs, 2015) which does indeed provide some useful insights related to developing a better understanding of MVPs. The key tenant of which, provided by Calabretta and Kleinsmann (2017), appears to suggest the most fundamental objective is to leverage resources and capabilities to make intangible insights, ideas and concepts tangible, sharable and understandable. Finally, while Shepherd and Gruber (2021) provide this fundamental attribute, which appears to be an important development, the remaining discussion represents only more questions and calls for future research as they note this merely represents a simple step forward in our understanding.

The MVP concept as a framework instead of a definition

In this section, we begin to formalize our ideas on a framework that better encapsulates the understanding of an MVP beyond the existing definitions presented in our literature review. Table 3 collects common themes of the definitions that are present in Table 1 and is the start of our work toward a framework for fully understanding the concept of MVPs. Specifically, all existing definitions in Table 1 define MVPs in terms of three specific categories of concepts: process-based, goal-based or element-based. Some definitions might define an MVP in terms of just one of these conceptual categories, for example, York and Danes (2014) state that an MVP is just a set of "minimal requirements" that meets the needs of early adopters. The definition is element-based in that it describes the elements that should be included (the minimal requirements that meet the needs of adopters) but does not describe the process for using the MVP and does not describe the

Table 3. Common MVP definition themes

Common process based themes	Common goal-based themes	Common element-based themes
1. Making something new	1. Generate revenue	1. Minimum features
2. Minimal effort/investment/time	2. High return on investment	2. Absolute minimums
3. Imperfection	3. Generate customer virality	3. Critical features
4. Deployability	4. Validate business model	4. Minimum feature sets
5. Test hypotheses (scientific method)	5. Solve customer problems	5. Minimal requirements
6. Perform validated learning (customer feedback)	6. Identify valuable features	6. Smallest implementation
7. Iteration (build-measure-learn)		7. Least possible
		8. Objects/artefacts/proxies

Source(s): Authors' own work

goal that the entrepreneur should be working toward with the MVP. Other definitions might address two or three of these categories, however, would still be incomplete since the full concept of MVP would encompass more than what a singular sentence-based definition could provide. Hence, our framework for a full MVP concept instead avoids proposing a definition in favor of a fuller understanding based on these categorical collections of common themes across definitions. In the sections that follow, we define these categories, present full descriptions and offer clarity in areas where there currently are only vague suggestions.

Process-based themes of the MVP concept

Process-based definitions of the MVP concept focus on the methodologies, procedures and workflows involved in developing, iteratingand refining a product. This approach emphasizes how the MVP is used as part of an ongoing process to gather feedback, learn from users and make incremental improvements. There are seven consistent themes across all definitions that include a process-based approach that are presented in Table 3.

Theme 1 of process-based definitions describes the process of using an MVP that consists of something new being created no matter how simple it might be (e.g. Lenarduzzi and Taibi, 2016a, b). In fact, the "minimum" nature of MVP does not just apply to the ideas of what is built or developed, but also speaks to Theme 2 of process-based definitions that highlight small amounts of investment in effort, money and time into the process (e.g. Lee and Geum, 2021). This second idea of limited investments into the process lead to Theme 3 of imperfection of the process due to the nature of the MVP that is being built (e.g. Miski, 2014). Specifically, that the process should not be perfect and that an entrepreneur should prioritize progress over perfection to get to Theme 4 quickly, or the idea of deployability (e.g. Read *et al*., 2014a). These ideas are consistent in process-based definitions that stress that an entrepreneur must put something in front of potential customers.

These first four themes of the process-based definitions lead to the last three themes of testing (e.g. Yli-Huumo *et al*., 2015b), validated learning (e.g. Ries, 2011) and iteration (e.g. Mikkonen and Systä, 2014). The process of using an MVP involves developing something new (Theme 1) quickly, easily and cheaply (Theme 2), which will not be perfect (Theme 3), and deploying that new thing to potential customers (Theme 4) for the first time. Then, the process of testing business model hypotheses according to traditional Scientific Method (Theme 5), to perform validated learning (Theme 6) takes over so that the entrepreneur can iterate according to the Build-Measure-Learn feedback loop (Theme 7). Arguably, these last three Themes of the process-based definitions can all be collapsed as they do argue for similar processes. Hypothesis testing as proposed by Blank and Dorf (2013) falls into the ideas of the scientific method in which an entrepreneur applies the ideas of hypothesizing how a customer will respond and gathering data to validate or invalidate those hypotheses. Ries' (2011) ideas of validated learning stress the same ideas with the emphasis on ad-hoc purposeful learning in contrast to post-hoc rationalization that can be encompassed into traditional scientific method of hypothesis, testing, analyzation and forming a conclusion. Finally, ideas related to the Build-Measure-Learn feedback loop (Theme 7) from authors such as Ries (2011) and Seufert (2014) are also consistent with traditional scientific method applications (Musante, 2009) that suggest a never-ending process that continues to iterate based on previous hypothesis testing in order to continue learning and the development of new knowledge.

Thus, our proposed framework, presented in Table 4, describes the concept of how MVPs are used with five distinct ideas. When using an MVP, entrepreneurs are deploying something new and imperfect with as little investment as possible to go through the scientific method of testing hypotheses and iterating based on testing results. Our contention would be that if an entrepreneur is doing something that falls outside of this described process, then they are doing something that most likely falls outside of the MVP concept.

Table 4. MVP conceptual framework

MVP process: how MVPs are used	MVP goal: what MVPs produce	MVP elements: what MVPs are
Making something new	*Required outcome: customer development*	*Required elements*
Minimal effort/investment/time	Customer discovery	Art
Imperfection	Customer validation	Distribution channel
Deployability	Customer creation	Customer feedback mechanism
Scientific method		
	Optional outcomes	*Guiding design ethos*
	Generate revenue	Lean essence
	Return on investment	
	Generate customer virality	

Source(s): Authors' own work

Goal-based themes of the MVP concept

Goal-based definitions of the MVP concept focus on the specific objectives or outcomes that the MVP aims to achieve. The goal-based approach is centered around the purpose of the MVP and what exactly the process-based aspects of the MVP concept should achieve. There are six consistent themes across all definitions that include a goal-based approach that are presented in Table 3.

Theme 1 and Theme 2 of the goal-based definitions suggest that there are tangible outcomes an entrepreneur might target with their MVP in terms of revenue generation (e.g. Poole, 2012) and a high return on investment (e.g. Erder and Pureur, 2016), however, these are not as common as other outcomes. In fact, many definitions ignore revenue generation altogether, meaning that return on investment might be thought of more from a return on investment in time or effort and not necessarily money. Theme 3 of generating customer virality through word-of-mouth and other customer recommendations (e.g. Poole, 2012) is also mentioned as a goal of an MVP in terms of how early adopters exposed to the MVP might then share it with other potential customers. We would contend that MVPs can make money, but that they do not have to. Instead, the MVP concept has other goals that are mandatory, whereas revenue generation is optional. Further, while it is beneficial if customers share the MVP for the startup, this might not be a requirement for all business models, especially if the entrepreneur is not assuming that they will use a viral strategy for customer acquisition. Therefore, while many definitions state these Themes as goals of an MVP, they do not present themselves consistently among the definitions as required outcomes and instead exist as optional outcomes that an entrepreneur might choose to pursue.

The next three themes that are more common of goal-based definitions are the goals of validating the business model (e.g. Kline *et al.*, 2013), solving customer problems (e.g. Moogk, 2013) and identifying valuable features customers would want in offerings (e.g. Järvinen *et al.*, 2014a). In effect, these types of goals in which the entrepreneur is successfully interacting with customers and receiving feedback that suggests that the MVP is validating business model hypotheses, solving early adopter customer problems and identifying key features for future iterations is very close to what Blank (2005) labeled as customer development. As Blank and Dorf (2013, p. 24) describe, customer development is the process through which startups use a structured process to iterate until they achieve, "'escape velocity' – enough measurable progress in finding the business model as defined by the board and team – to propel forward" through each of the four steps of customer development. Thus, these common goal-based themes of MVP's definitions all describe the goals of the customer development process: customer discovery, customer validation,

customer creation and company building. In essence, the MVP concept can be simplified by simply stating the goal of any MVP is to complete the customer development process.

Therefore, as we present in Table 4, our MVP Conceptual Framework proposes that the goal of any MVP should be customer development in the form of discovery, validation and creation. All MVPs have the goal of validating a business model that solves a customer problem and iterating to a point where eventually customer demand transforms the startup from a temporary search organization into an established business (Blank and Dorf, 2013). If the MVP also generates revenue or virality among customers, then that is an added benefit of using the MVP, however, entrepreneurs will always be using the MVP to focus on the goal of discovering, validating and creating customers.

Element-based themes of the MVP concept

The final category of themes that is common among MVP definitions is the descriptions of what MVPs are, or should be, in terms of the elements that make them. Element-based definitions of MVP focus on the essential features, components or attributes that constitute the MVP. Definitions that include descriptions of elements attempt to emphasize what the MVP is, describing the minimum set of features or functionalities necessary for the product to be viable to carry out the process of using the MVP to accomplish the goal of the MVP.

As it can be seen in Table 3, there are eight different common themes of element-based definitions, however, all eight of these themes collapse into two large ideas. First, MVP definitions often stress the minimum, minimal and/or smallest implementation of features, feature sets or requirements (Themes 1 through 7). Second, element-based definitions then offer that MVPs are objects, artefacts or proxies, and potentially could be some combination of all three. In reviewing the literature, definitions and discussions that highlight the elements that make up MVPs are the most problematic in the sense that there is no clear understanding of what the required elements are of an MVP outside of these two ideas. That is not to say that these starting ideas are not useful or correct, we simply view them as incomplete. Many authors point to the minimum nature of an MVP, but there is not much to guide entrepreneurs in terms of what exactly those elements might consist of given where they are in their specific process. So, we take from these existing ideas and definitions a starting point for our contribution: that MVPs are *something* in terms of an object, artefact, or proxy and that entrepreneurs should make those things as minimally as possible at the point in their specific MVP process.

The recent discussion by Shepherd and Gruber (2021), based on the prototyping literature, highlights that a prototype is below what is minimum if it fails to make intangible insights, ideas and concepts tangible, sharable and understandable to hypothesized stakeholders. If a prototype does make a potential opportunity tangible, sharable, and understandable and stakeholders reject it, then the prototype is an MVP and has served its purpose as a vehicle for learning. Obviously, stemming from the prototype literature, a tangible prototype is particularly relevant. However, in terms of developing an MVP, which may or may not take the form of a prototype, the element of tangibility is not essential as MVPs may be sketches, designs, simulations, or experiments all of which may or may not have physically tangible elements. Thus, the two essential characteristics that comprise any MVP element appear to be shareability and understandability, though certainly tangibility will commonly contribute to understandability. Therefore, both shareability and understandability are necessary for any MVP to fulfill its purpose and any individual element that goes into creating an MVP should be assessed on how that element helps in making the MVP understandable or shareable.

We also adopt the perspective that an MVP is a boundary object that should be purposely designed to enable communication across a boundary. Regarding an MVP, the important and meaningful boundary which is often challenging to bridge is that which resides between the new venture/founders and the intended customers (Nonaka, 1994). Boundary objects should serve as a bridge that allows for, and promotes the transfer of, information across the boundary

thereby reducing the time required to learn (Grichnik *et al*., 2016). Therefore, we also endorse Shepherd and Gruber's (2021, p. 979) sentiment that an MVP should be an effective boundary object capable of "providing (1) a shared language for two parties to represent their knowledge to each other, (2) a means by which both parties can learn about their differences, and (3) a process by which they jointly transform their knowledge (Carlile, 2002) in a way that enhances learning about the focal startup."

Though the key attributes of shareability and understandability are paramount in terms of developing either a prototype or an MVP, the question remains, what enables an MVP to achieve sufficient levels of shareability and understandability? First, we argue it is evident that some sort of representation of a version of the product/service must be included. Further, this version, while it does not need to be perfected to any degree, must deliver the key and most fundamental value proposition(s) associated with the product/service (Lortie *et al*., 2022; Winkel *et al*., 2020). Second, there must be some mechanism for getting this version of the product/service in front of the customer. Specifically, there must be a way to present potential customers with a version of the product. Third, and finally, there must be some mechanism for the customers to provide feedback about this version of the product/service. Thus, we have identified three distinctive elements which we propose are the fundamental elements necessary for something to be considered an effective and complete MVP. Put more succinctly, the three elements we have identified are as follows: (1) art, (2) distribution channel and (3) user feedback mechanism. The following sections provide definitions and explanations about each of the three fundamental categories of elements that are essential for any viable MVP along with examples of elements that fall into each of these three categories.

Art as a required element of an MVP. In defining art, we refer to the United States Patent and Trademark Office's (USPPTO's) (2023) broad definition of prior art which refers to any evidence that an invention is already known or publicly available before the filing date of a patent application, including published patents or patent applications, published articles or academic papers, public documents, public demonstrations or presentations, and prior use or sale of the invention. Thus, art may take the form of a prototype that is tangible or simply a digital manifestation of essential elements of the willful design (Lauff *et al*., 2018). This art should represent a design made before final artifacts exist and are not production stage design (Buchenau and Suri, 2000). This iterative tool requires the minimal amount of time, energy, and money to be used for learning. Specifically for the purposes of the MVP, the prototype serves as a boundary object for stakeholders to interact with and to facilitate communication (Shepherd and Gruber, 2021). The art can come in a variety of forms, each with their own set of benefits and/or disadvantages.

Currently, art is generally placed into two format categories, physical or digital embodiments (Lauff *et al*., 2018) though even broader conceptualizations may be possible. Both art forms have their unique benefits, the type of distribution channel, desired feedback and customer segment will determine which form is most viable. The physical art deployed in an in-person distribution channel stimulates the customer by heightening the amount of tangibility present. The physical art in the in-person distribution channel leads to increased interaction and input by the customer (Deininger, 2018). Traditional prototyping from the design thinking perspective uses this approach frequently, as prototyping seeks more qualitative feedback on early design concepts that may not be entirely formulated yet. Physical prototypes and other forms of art lead to increased sharing of stakeholder opinions and newly discovered design requirements that the stakeholder may introduce (Deininger, 2018). A physical prototype can also be used in a digital channel, such as on a landing page. Art can of course be digital, or non-physical in nature. Non-physical art takes the form of any virtual embodiment of a design concept that does not include a real image or video of a physical prototype. CAD (Computer Aided Design) renderings are included as virtual prototypes. The factor that differentiates the two formats is if the customer believes that the current iteration of the prototype can be physically touched if purchased. While tangibility is still present due to

the visualization of the physical art, it is decreased due to the virtual nature of viewing the prototype through a digital distribution channel.

Art within the context of an MVP can exist as a variety of types that each have their own traits that can be used for a specific learning outcome. More specifically, all MVPs require some evaluative element which can be presented, assessed, interacted with, and/or evaluated by potential customers. Typically, these elements may consist of either direct interaction with a version of a prototype, or a visual assessment of a potential, or envisioned prototype or description. This establishes the previously discussed fundamental and essential attributes of shareability and understandability. Therefore, we have assessed the most common art elements (e.g. Osterwalder *et al.*, 2014; Petrakis *et al.*, 2021) in terms of their level of shareability as well as their understandability. The results of this analysis are presented in Table 5 that provides a 3×3 framework consisting of each element.

Table 5 provides interesting insights on the fundamental viability associated with distinct types of MVPs which allow for customer shareability and understandability. Most insightful and striking is the fact that no viable currently known art elements reside in the top left region of the 3×3 depiction. Certainly, it might be expected that elements that are characterized as having low levels of shareability and understandability could not be considered viable as MVPs, which we do find. This is reasonable as the ability to gather meaningful customer feedback would be insufficient to be considered an art element. However, more interesting, is that no combination of low shareability or understandability and medium of either appears to qualify as a viable art element. Put simply, no combination of low and medium attributes appears to be viable in terms of generating meaningful and useful art. Thus, we have an interesting and important resulting framework suggesting that either one element (i.e. shareability or understandability) must be considered high, or that both attributes must be considered medium in terms of capturing customer assessments. At the same time, it is worth noting that simple MVPs can be developed with basic levels of art that do not necessarily have high levels of understandability or shareability. For example, an MVP that makes use of text alone for the art element would technically meet the requirement of representing a final design and could be written, or even verbal (in the sense of an entrepreneur explaining their final product or service idea to a user).

It should also be noted that different art elements lend themselves to the nature of early-stage testing when an entrepreneur is just getting started to stay within what we refer to as the "Guiding Design Ethos" of a Lean Essence in Table 4 that shows our MVP Conceptual Framework. While certain art elements create more understanding and shareability, they may also represent more expensive or time-consuming developments. Therefore, an entrepreneur may choose simpler art elements earlier in the process and move to more advanced art elements in later MVP iterations.

Table 5. Assessment of art

		Level of shareability Low	Medium	High
Level of understandability	Low	N/A	N/A	Sketch Images Book
	Medium	N/A	Text Video Wireframe	3D prototype
	High	Service Roleplay/storytelling	Physical product Physical model	Application

Source(s): Authors' own work

Distribution channel as a required element of an MVP. The next required element of any MVP is the ability to present the art to potential customers. This becomes necessary based on how we conceptualize our framework of how MVPs are used and what they produce and accomplish. We explained how various attributes are likely to allow for shareability and understandability based on the art elements described above. However, the question remains as to how the MVP is presented to customers prior to considering how customer feedback is conveyed or captured. Put simply, an MVP must be capable of being distributed, by one mechanism or another, to potential customers, otherwise it will not be viable and must be determined insufficient. Conceptually, an entrepreneur that has an art element, but does not have a way to put the art element in front of a customer, is not able to accomplish the process or goal of an MVP.

Distribution channels within the context of an MVP may exist in a wide variety of forms. Each has their own traits that can be used for a specific learning outcome. Distribution channels are critical to test an MVP's market potential and gain user feedback. In Table 6 we evaluate the attributes of these distribution channels by ranking their shareability and understandability. Shareability is defined as the ability to share with customers. Understandability is the ability to grasp the purpose of and interact with the MVP while incorporating the various art elements an entrepreneur has chosen to utilize.

Certain distribution channel elements may be more appropriate for getting art elements in front of potential users when considering their shareability and understandability. The distribution channel may be considered insufficient if one of these two attributes is lacking and we acknowledge that many MVPs might in fact make use of multiple distribution channel elements in combination. For instance, a direct mailing campaign may be able to distribute art elements with hundreds of potential consumers (i.e. shareability) but the ability to grasp and fully understand the product/service offering will typically be low. It is not enough to simply distribute an MVP if the distribution channel ranks low on shareability and understandability and will not allow the entrepreneur to best show the art that has been developed.

Like art elements, our conceptual framework of MVPs highlights that the guiding design ethos for distribution channels should be a lean essence in which the MVP process should always be followed. Minimal effort, investment and time should be allocated toward developing distribution channel elements and the entrepreneur should follow the iterative nature of scientific method according to the MVP conceptual framework of how MVPs are

Table 6. Assessment of distribution channels

		Level of shareability Low	High
Level of understandability	Low	Flyers Physical mail Online ads Print ads Phone calls Ads in multi-sided	Landing page Crowdfunding page Email Social media Text message Mail Television Streaming services
	High	In person In app purchases Pitch presentation Office meeting Pitch to colleague	Phone app stores Retail MLM Video game platform Multi-sided platform VR META

Source(s): Authors' own work

used. This guiding idea supports the ways in which entrepreneurs can lower risk and uncertainty while using an MVP.

User feedback mechanism as a required element of an MVP. The third essential requirement of any MVP is the ability for customers to provide feedback to the entrepreneur as can be seen from the MVP conceptual framework in Table 4 based on the process and goal. There are many mechanisms for capturing feedback that vary in terms of options that founders might adopt to capture essential feedback from users. Regardless of whether the feedback is in the form of an action to be performed (e.g. providing an email address to sign up for a waiting list or perhaps making an early purchase in the form of a pre-sale), more detailed qualitative assessment (user evaluation and feedback via interview), or more strictly quantitative (number of views or watch time), some form of feedback must be considered an essential element of an MVP. A piece of art distributed via a distribution channel, without a viable feedback mechanism in place, cannot be considered a complete MVP as the user would have no way to communicate back to the entrepreneur their thoughts, likes, dislikes, suggestions, intentions to purchase, etc.

Unlike the first two categories of MVP elements in which understandability and shareability are characteristics of MVP elements, the user feedback mechanisms flip the direction of information flow for these characteristics. Users interacting with the MVP utilize the feedback mechanism to communicate back to the entrepreneur their understanding and the potential shareability of the MVP or business idea. For example, certain user feedback mechanisms are better than others when it comes to allowing for the user to communicate back to the entrepreneur how well they understand the idea behind the MVP and the product or service idea that the MVP represents. Some feedback mechanisms may be extremely limited in how well the user can communicate their understanding of the product or service idea if the user feedback mechanism is more quantitative versus qualitative. For example, the ability for users to communicate their understanding of the MVP back to an entrepreneur that designs their MVP in a way where the intended meaningful action for the user is to submit an email to sign up for a waitlist will be inherently lower than other feedback mechanisms. This happens both when users sign up, but also when they do not sign up. First, a user that does sign up is limited in providing information, feedback or data when all they can do is submit their email for sign-up. This is further limited when the user does not submit their email since the entrepreneur receives nothing at all if this is the only feedback mechanism that has been built into the MVP. This email sign up could be contrasted with a survey that the entrepreneur could implement either on its own or in addition to the email sign-up that could allow for the user to communicate their understanding of the MVP in much more detail.

Like the way user feedback mechanisms allow users to communicate their understanding back to the entrepreneur, user feedback mechanisms also allow for users to communicate the shareability, or potential shareability of the product or service idea, back to the entrepreneur. In this case, the shareability of the potential product or service is communicated back to the entrepreneur via the meaningful action of the user feedback mechanism based on the nature of the feedback mechanism itself. For example, imagine that an entrepreneur is attempting to make use of a user feedback mechanism in the form of a pre-sale. The meaningful action of the user participating in the pre-sale is a small level of feedback that the user might also be interested in sharing the MVP or the actual product with others. And the user might also do exactly that via methods that are not inherently trackable to the entrepreneur if the user uses offline word-of-mouth to let colleagues know about the MVP and pre-sale, they just participated in. However, an entrepreneur could also make use of a user feedback mechanism in which another meaningful action that the user can take is to provide emails of people that they would like to share the pre-sale. This meaningful action would not only allow for the entrepreneur's MVP to have more virality, but it would also allow for users to communicate directly to the entrepreneur how willing they are to share the MVP or the product or service the MVP represents. In fact, based on the design of the MVP and pre-sale user feedback mechanism, the entrepreneur would have a highly trackable measure of the shareability of the MVP and could calculate a viral coefficient (Croll and Yoskovitz, 2013) based on the ways in

which users provide emails of those they would like to share the MVP with. We formalize these ideas by presenting and assessing a typology of potential user feedback mechanisms based on their abilities for users to communicate back to the entrepreneur their understanding and potential shareability in Table 7.

An important note should be mentioned about user feedback mechanisms and how entrepreneurs design their MVPs to allow users to communicate back their level of understanding and the potential shareability. While the user feedback mechanisms presented in Table 7 consider meaningful actions that will allow the entrepreneur to receive varying levels of feedback about how well the user understands the MVP and how willing they might be to share it, the table does not necessarily consider the commitment, investment or "ask" that the feedback mechanism requires. This is certainly something that entrepreneurs inherently balance in the design of their MVPs since if a user feedback mechanism is asking for too much from the user, then the user will simply choose to do nothing and not make use of the feedback mechanism at all. For example, an entrepreneur in the early ideation stage of testing might choose to make use of a feedback mechanism that has a much smaller ask since at this stage any feedback is better than nothing. So, this entrepreneur might make use of a survey since it can be designed to be quick, easy and free for the user to provide a large amount of feedback on understandability and potential shareability over say sharing with friends or making a purchase (two feedback mechanisms that are a much higher ask on the part of the user). Again, lean essence is the guiding design ethos for any design choice entrepreneurs make about MVP elements.

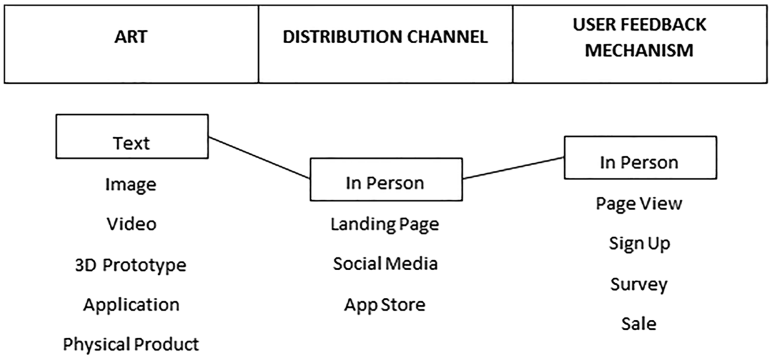
Building an MVP – an à la carte approach. We now present a practical representation of how entrepreneurs can build different types of MVPs, whether they be the simplest incarnations to more advanced MVPs, given the MVP elements we have presented above. Blank and Dorf (2013) discuss the difference between low-fidelity and high-fidelity MVPs, which differ largely based on the functionality of the MVP. Low-fidelity MVPs are quick, cheap and largely non-functional representations of the product or service while high-fidelity MVPs can take more effort to develop, may have more significant costs, and will start to introduce functionality via working prototypes or beta versions (Blank and Dorf, 2013).

Figures 1–3 present truncated lists of art, distribution channels and user feedback mechanisms curated from our ideas in Tables 5–7. Our contention is that given these options, an entrepreneur is largely free to pick and choose from these lists as they see fit to follow the MVP process toward the MVP goal. In addition, entrepreneurs may also choose multiple elements from the same category if inclined. Progressively, from Figures 1–3, we present the most basic of MVPs that an entrepreneur may choose to an advanced MVP build progressing along the ideas of low-fidelity to high-fidelity.

Table 7. Assessment of user feedback mechanisms

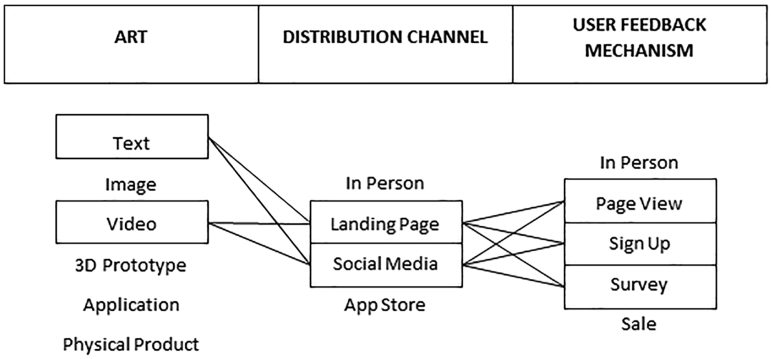
		Ability of user to communicate shareability	
		Low	High
Ability of user to communicate understanding	Low	Sign up Video view Page view Read Posting	Email referral
	High	Sales Pre-sales	In person Survey Focus group Investment

Source(s): Authors' own work



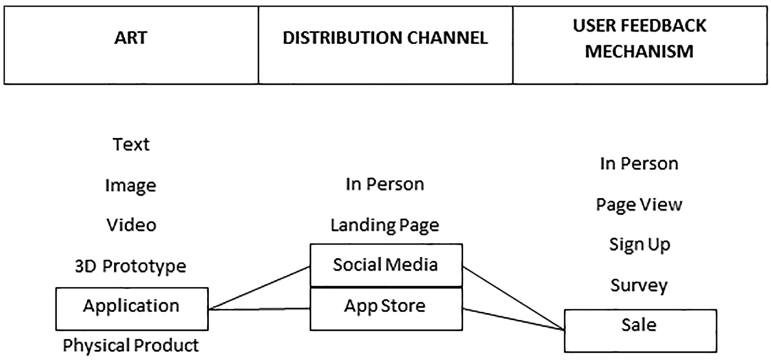
Source(s): Authors' own work

Figure 1. Simple MVP build example



Source(s): Authors' own work

Figure 2. Moderate fidelity MVP build



Source(s): Authors' own work

Figure 3. High fidelity MVP build

Figure 1 depicts a simple MVP, one that represents Blank and Dorf's (2013) "get out of the building mantra" that starts the customer development process. In effect, this would be the lowest fidelity MVP possible in which the entrepreneur needs nothing other than their own time and an idea of the conversation they want to have with potential users. In this case, the art the entrepreneur makes use of is their own spoken text, distributed in person to a potential user, and a user feedback mechanism of the user responding back in person. In effect, a basic conversation or customer interview. Figure 2 depicts a more advanced MVP build in which the entrepreneur has chosen written text and video as their art, a landing page and social media for their distribution channel, and page views, sign ups, and a survey as user feedback mechanisms. While this MVP build might allow for larger distribution of perhaps a video that shows more of the product or service than spoken text could, it would also take more time and effort to build compared to the MVP in Figure 1. In addition, Figure 2 shows that the user can provide feedback simply via web metrics that would track the number of page views, signing up on the landing page, and/or filling out the survey the entrepreneur has designed. Finally, Figure 3 presents a high-fidelity MVP build in which the entrepreneur has chosen to use a working application for art, distributed via an app store and social media, and a user feedback mechanism of a sale on that app store. This high-fidelity MVP would have working functionality but might not be the final version of the application with all the features that the entrepreneur intends for the final product offering.

As noted above, Figures 1–3 show that an entrepreneur can create any MVP they desire if they had a minimum of one element from each category. MVPs can take various levels of functionality and looks depending on the creativity of the entrepreneur and the business model they are attempting to test. More advanced MVPs can be built by combining multiple elements from the same category; however, this is not required as a single element from each category would create an operational MVP. Again, we highlight the guiding design ethos of choosing MVP elements of lean essence should be followed when making design decisions.

Conclusion and implications

We sought to develop an explicit and appropriate framework for understanding the most fundamental and essential elements of the MVP. In doing so, we reviewed and drew insight from earlier definitions and conceptualizations of MVPs from across both the academic and practitioner literature. We also drew heavily from state-of-the-art efforts to bridge the divide between the two (e.g. Shepherd and Gruber, 2021). We then presented our MVP conceptual framework that consists of the MVP process, MVP goal and MVP elements. While previous understanding allows for good understandings of the MVP process and MVP goal, we further sought to elaborate on MVP elements. Stemming largely from ideas within prototyping literature, we discerned that fundamental essential attributes any MVP will require include understandability and shareability. Further, we developed actionable and real-world options for the elements that will result in a viable MVP.

Our efforts lead to the development of a more detailed and comprehensive framework defining the nature and essential elements required of any MVP. Those elements included the visual representation of the product/service that delivers the value proposition – art, the ability to present this version of the product/service to the customer – distribution channel, and a method for gathering customer opinions and evaluation – user feedback mechanism. All of which will vary, depending on the business model, in terms of their shareability and understandability.

In our view, this presents an important contribution to the existing literature related to both business models and more specifically MVPs as they relate to business models. Before this detailed comprehensive framework, explanations around the development and deployment of MVPs were limited by the lack of definitional clarity, general vagueness, and conflicting ideas and requirements. Put simply, one could not easily determine what was genuinely an MVP and what was not along with how they might use an MVP and for what purpose. Nor could one be fully informed as to which course of action would inevitably result in a useful MVP. We are hopeful that this framework will be beneficial to future researchers, practitioners and

educators. The MVP framework we have developed here lays the groundwork for a more systematic and comprehensive understanding of MVPs, offering valuable insights for future research. However, several areas still require further exploration to enhance the utility and applicability of MVPs in various entrepreneurial contexts.

Implications for future research

Employing the MVP framework developed here should prove to be immensely useful for future research, enabling more empirical work and subsequent analysis related to diverse types and variations of MVPs and their effectiveness. This framework allows for classification, measurement, categorization and evaluation of distinct types of MVPs, which has historically been underdeveloped despite the widespread popularity of this approach to entrepreneurship and business model assumption testing and validation. Some MVPs may prove to be effective for certain products or services and business models, whereas others might not. Currently, it remains unclear which MVPs would be most appropriate for differing ventures. Future research should work toward developing a more comprehensive theory that integrates the elements of an MVP explained here, along with contextual factors, external influences and likely outcomes associated with developing MVPs.

Deepening the understanding of MVP applications is another critical area for future research. While our framework provides clarity on the fundamental elements of MVPs, there is still a need to explore the specific conditions under which different types of MVPs yield the most valuable insights. For instance, MVPs may perform differently in markets characterized by high uncertainty versus those with established customer preferences. Future studies could examine how variables such as market dynamics, customer segment diversity, and technological innovation influence the efficacy of different MVP configurations. This research could also address the gap identified in the literature regarding the contexts where MVPs might fall short, thereby extending the theoretical foundations of MVP usage.

Integrating MVPs with broader theoretical frameworks, such as effectuation theory or dynamic capabilities, represents another promising avenue for future research. This synthesis could help address some of the limitations of MVPs by providing a more robust framework that accounts for the dynamic and multifaceted nature of entrepreneurship. For example, combining MVPs with effectuation theory could offer insights into how entrepreneurs navigate uncertainty and leverage available resources in the early stages of venture creation. Additionally, research could explore how dynamic capabilities influence the iterative process of MVP development, particularly in rapidly changing markets. These extensions could lead to a more holistic understanding of how MVPs function within the broader landscape of entrepreneurial activities.

Future research can also shed light on how entrepreneurs might best navigate the balance of resource commitment required for MVPs, as minimum resource commitment is indeed a part of developing an MVP, as our framework shows. Firms with more resources will inevitably be able to leverage these resources and capabilities to generate more detailed and comprehensive responses from the external environment. For example, resources can be leveraged to create, establish, and appropriately manage various communication channels and mechanisms, thereby enabling more efficient processes for both generating and capturing feedback in response to MVPs. In contrast, firms lacking these capabilities may struggle to generate sufficient responses. Interesting future research might determine the ideal number of resources that need to be committed to testing specific hypotheses. Some research has already begun to identify optimal methods for developing MVPs (e.g. Lortie *et al.*, 2022).

We also highlight the contribution our framework makes to understanding MVPs from a definitional standpoint. As we have argued, no single definition can fully encompass the MVP concept, and therefore, future research should be cautious in attempting to simplify discussions around MVPs in such a way. Instead, we suggest that our framework be utilized and further built upon in future research to achieve a complete understanding of what MVPs are, what they do, and what they accomplish.

Implications for entrepreneurship educators

Educators are expected to support and lead entrepreneurs with sound guidance. Building a proper MVP and defining its characteristics is a significant point in the process to understand how to assist entrepreneurs. If an educator cannot define an MVP, how is the educator properly serving his/her entrepreneurs or students? There is ambiguity in starting a venture and there are variables that cannot be controlled, customer behaviors as an example are not predictable with absolute certainty. However, it is under the educator's control to understand and relay the requirements of an MVP to provide the best chance of success to their entrepreneurs/students.

This paper has the potential to guide a classroom and provide structure to an entrepreneurial curriculum. If an educator assigns students to create a business idea in one semester, they may use this paper to guide their students in the MVP development. The educator may have a profound impact continuing teaching entrepreneurship past the concept stage and advancing his students toward MVP development. This comprehensive curriculum would gain notoriety within the institution and potentially beyond once students act upon this guidance to create an MVP and maybe one day a successful company. We envision this type of pedagogical approach being highly appropriate across many types of curriculums. For example, these concepts could be used in high school, higher education, co-curricular and community-based programs with students of any age. Truly, we believe, our conceptualization of MVPs in this paper are appropriate for any student learning about the entrepreneurial process. The educator may assign tasks and milestones to reach based on each required element.

In a capstone class for instance, the challenge could be to develop the business concept with an MVP using the elements provided in our paper as a clear guide. This is adaptable to different learning styles. An educator may incorporate any or all elements of the MVP guidelines to gauge their student's growth. More emphasis can be placed on using multiple art elements for those who are heavily creative, or priority can be given to the functionality of the MVP with a focus on feedback mechanisms and channels for those who are analytical and learn with data. Ultimately, we believe that entrepreneurship students of any type benefit from an experiential approach and this paper provides a conceptualization of MVPs that allows for the educator to create easier assignment and learning outcomes for their students when tasking them to come up with and implement MVPs in experiential exercises.

With this paper as an outline, we believe the concepts continue past formal training, as educators can practice what they preach by having a tangible impact outside of the classroom. Students can use this experience to launch their own startup after graduation or the formal learning environment they might be exposed to these ideas. It sets a firm entrepreneurial foundation that cannot be taken away from the student experience and can be called upon in future ventures years after training.

Implications for practitioners

Defining the full framework for the MVP concept along with specific MVP elements allows for clearer results, quicker testing, and provides a direct roadmap for entrepreneurs. Entrepreneurs need to test various hypotheses that are within their business model. If the founders can determine early on what to build as part of an early representation of the product or service, they will be more confident in their decisions. The founders will know they are collecting the proper amount of testing allowing them to mitigate risks and carve the best path for their ventures moving forward. It should be noted that there are risks for practitioners utilizing MVPs since they deploy incomplete ideas to the environment and potentially lack intellectual property protections due to resource constraints. If founders receive mixed definitions, and unclear guidance on what constitutes an MVP, they may over or under develop their art and may not consider the distribution channels or feedback mechanisms as part of this process. The results could be catastrophic, causing more money and time spent on building an MVP that does not satisfy minimum requirements to be validated leading to a failed venture.

We do not claim that MVPs are not without risk, however, we do believe our framework provides a way for entrepreneurs to make fewer mistakes and move through the process quicker to reach the MVP concept goals.

Designing pass/fail experiments with an MVP are meant to be short, simple and objective, not involving copious amounts of money or time. Failing may be inevitable regardless of properly building an MVP meeting all requirements. However, failing early with an adequate MVP saves the founders time and priceless resources which are commonly significantly constrained to pivot or direct their energy toward another segment. Defining an MVP aids the entrepreneur with next steps helping to advance the concept to a profitable business ultimately leading to product market fit with a clear value proposition (Osterwalder *et al*., 2014).

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